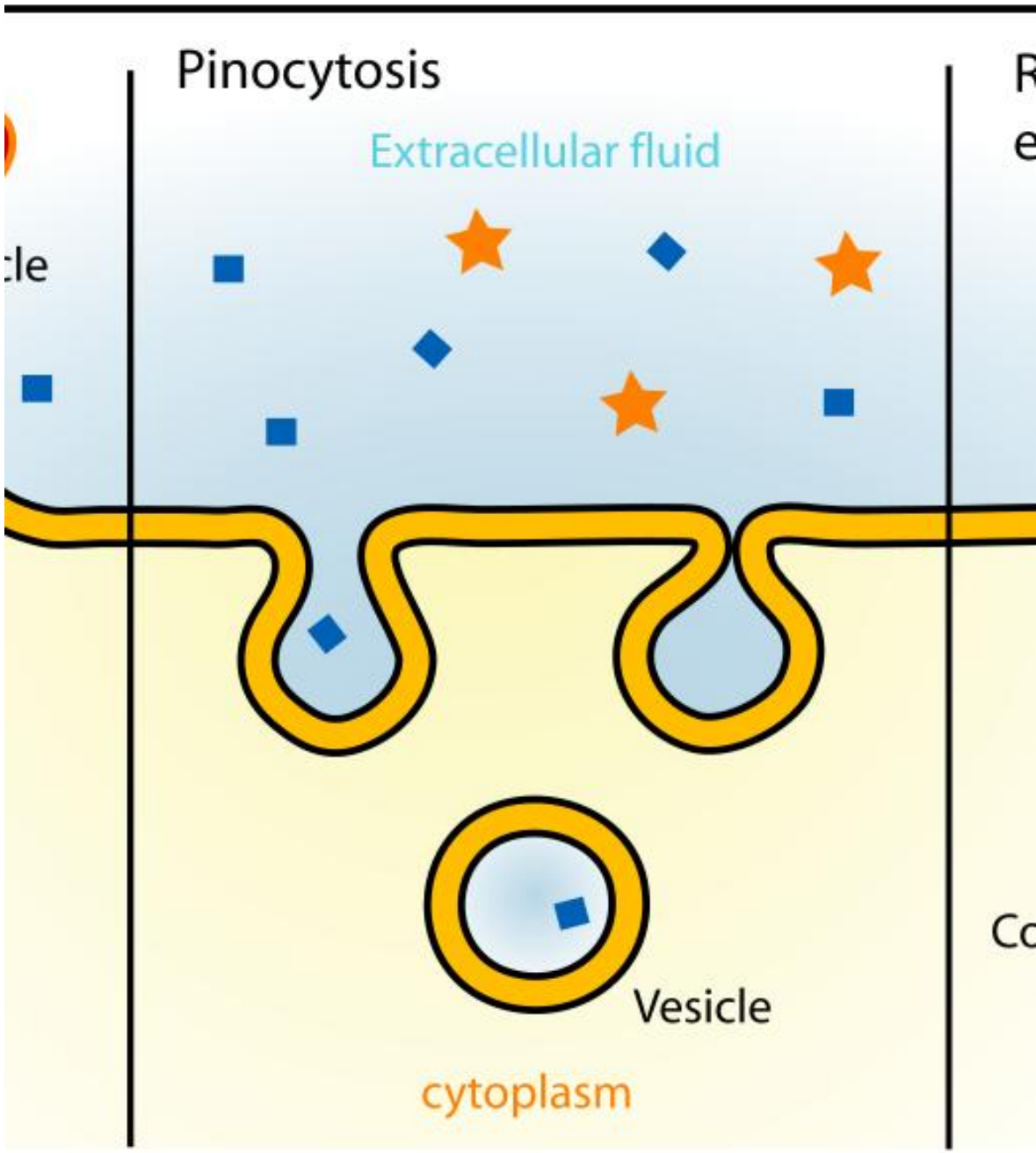
DP Biology – B2.1 (HL extension)

Membranes: fluidity, gates and pumps

- How do lipid and protein molecules come together to form membranes?
- What decides whether a substance can cross a membrane?



HL statements B2.1.11–17. Textbook pp. 237–243.
Image: three kinds of endocytosis. Mariana Ruiz Villarreal, public domain.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B2.1.11–13	Explain how fatty acids and cholesterol control membrane fluidity, and why fluidity allows vesicles to form and fuse
2	B2.1.14–15	Compare voltage-gated and neurotransmitter-gated channels, and explain the sodium-potassium pump
3	B2.1.16–17	Explain indirect active transport of glucose, and how cell-adhesion molecules hold tissues together

The big picture: a living, moving boundary



The membrane is **fluid**, so it can bend, bud and fuse. Its proteins **gate**, **pump** and **hold cells together**.

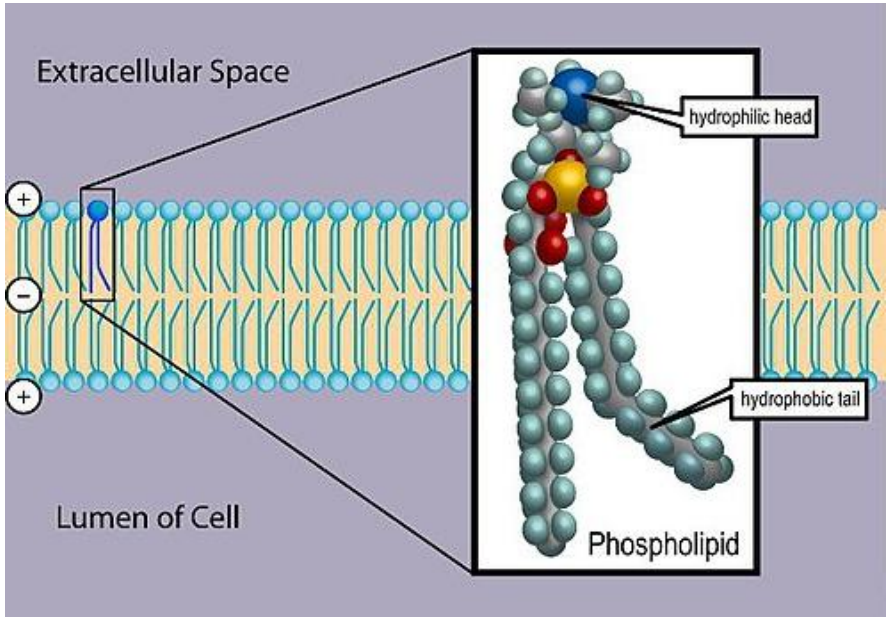
Big question

How can a cell absorb glucose against its gradient with a protein that uses no ATP?

Fatty acids and membrane fluidity

B2.1.11

	Saturated tails	Unsaturated tails
Shape	Straight	Kinked at C=C
Packing	Close	Looser
Melting point	Higher	Lower
Membrane	Less fluid	More fluid in cold

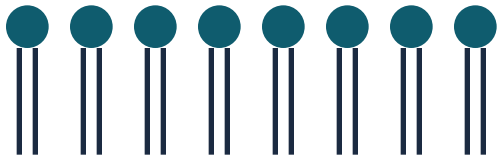


A phospholipid in a membrane. T. van Brussel, free use.

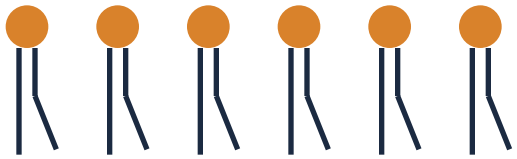
Key idea

Fluid enough for proteins to work; not so fluid that it leaks.

Why kinks matter



saturated: tight packing



unsaturated: kinks keep them apart

So

More unsaturated tails = weaker attraction between tails = more fluid membrane, even at low temperature.

Adapting membranes to temperature

B2.1.11



Lake sturgeon. Ryan Hagerty, USFWS, public domain.

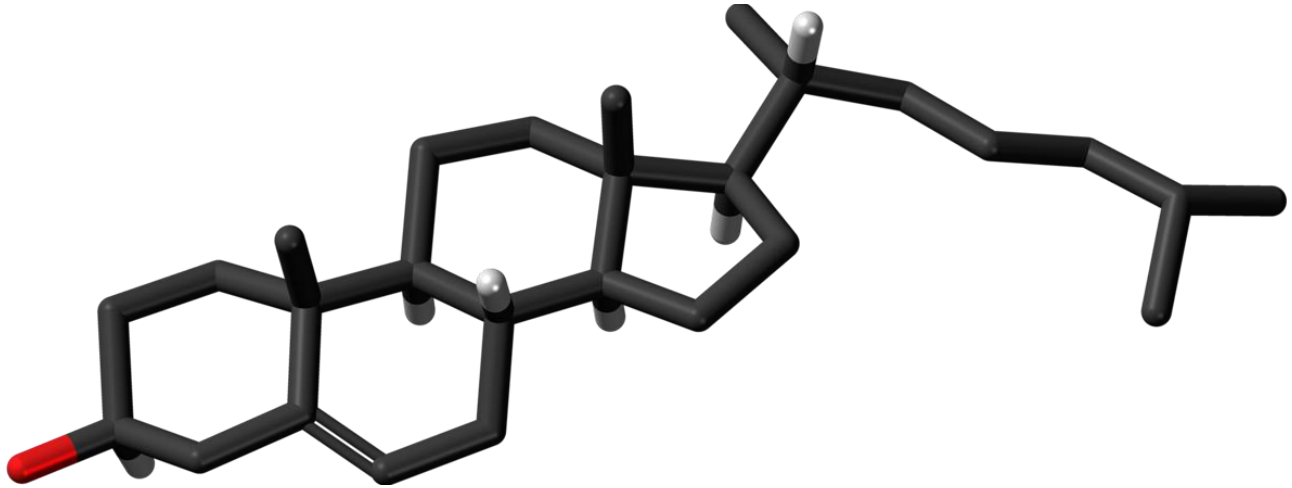
Cooled from 16 to 1 °C, sturgeon raise the share of **unsaturated** fatty acids in their membranes, keeping them fluid: **homeoviscous adaptation**.

Predict

How would membranes of a hot-spring bacterium differ from those of an Arctic fish?

Cholesterol: a fluidity buffer

B2.1.12



Cholesterol: red atom = oxygen of the -OH group. Jynto, CC0.

Part	Position
-OH (polar)	By the phosphate heads
Rings and tail	Among the fatty acid tails

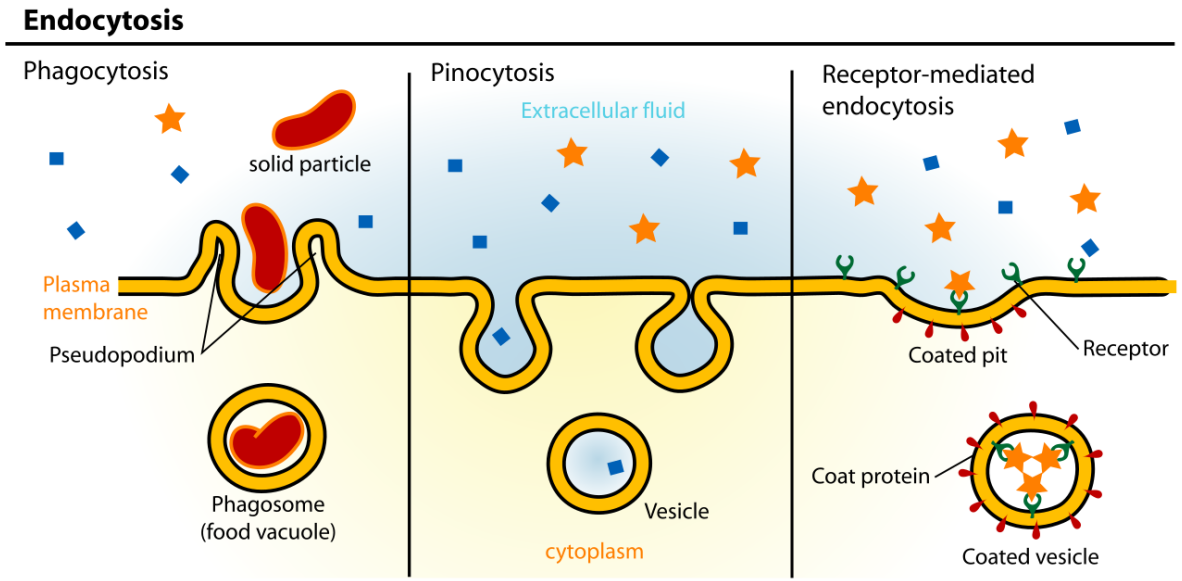
Temperature	Cholesterol
Low	Keeps tails apart: stays fluid
High	Restricts tails: less fluid

Common mistake

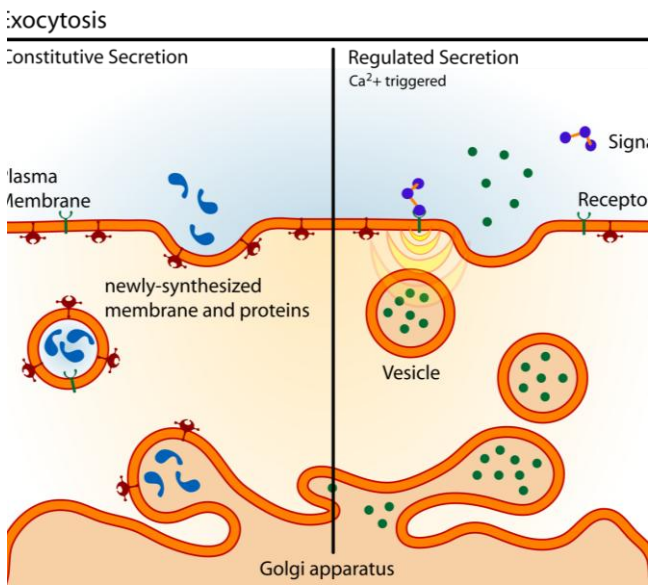
Fluidity is not permeability; cholesterol also reduces ion leaks.

Fluid membranes make vesicles possible

B2.1.13



Endocytosis. M. Ruiz Villarreal, public domain.



Exocytosis. M. Ruiz, public domain.

Both need **ATP**.

	Endocytosis	Exocytosis
Direction	Into the cell	Out of the cell
Membrane	Pinches inwards	Vesicle fuses
Example	Macrophages engulf old cells	Plasma cells secrete antibodies

Membranes on the move

B2.1.13

Explain

Why fusing leaves no hole.

Predict

Surface area during exocytosis and endocytosis.

Suggest

How a secreting cell keeps its membrane the same size.

Hint

Both are bilayers that can flow and join into one.

Working with data: fish membranes and temperature

B2.1.11

Unsaturated fatty acids in membranes of fish kept at different temperatures. Invented values; pattern matches real studies.

Water / °C	Unsaturated / %
5	68
10	61
15	55
20	48
25	42

1. Describe the trend.
2. Explain the advantage in cold water.
3. Predict cholesterol in animals from hot places.

Check-in: class 1

B2.1.11–13

1. Explain how unsaturated fatty acids affect fluidity.
2. Explain how cholesterol buffers fluidity.
3. Define homeoviscous adaptation.
4. Distinguish endocytosis and exocytosis.
5. Explain why vesicle fusion needs a fluid membrane.

How to use this

Answer without your notes, then check against slides 3–10.

Gated ion channels in neurons

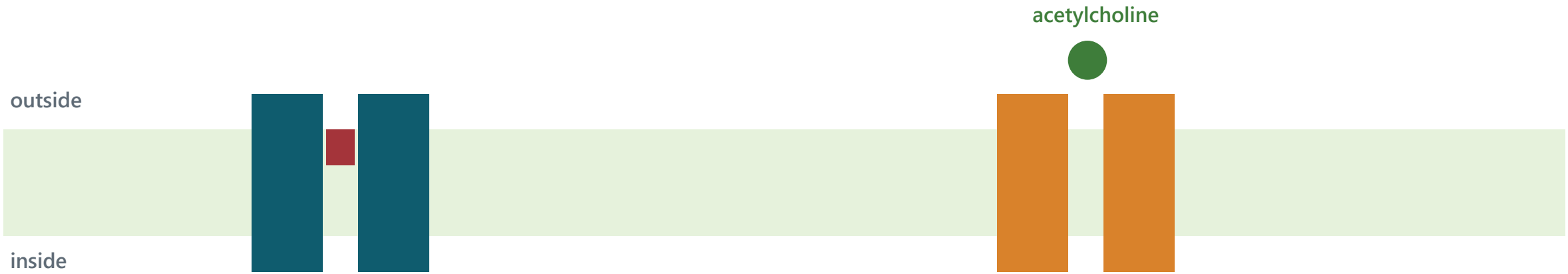
B2.1.14

	Voltage-gated	Neurotransmitter-gated
Opened by	Change in membrane potential	Neurotransmitter binding
Example	Na^+ and K^+ channels	Nicotinic acetylcholine receptor
Where	Along the axon	Post-synaptic membrane
Transport	Facilitated diffusion	Facilitated diffusion

Ions flow down their gradients, so channels use no ATP.

Two kinds of gate

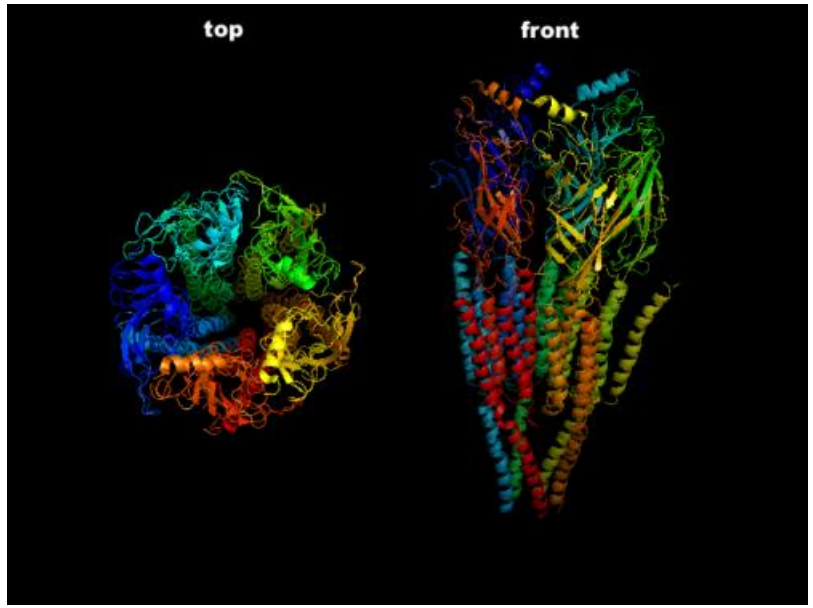
B2.1.14



K^+ channel inactivation

After opening, a ball of protein on a chain plugs the pore until the membrane repolarizes.

Two gates in action



Nicotinic acetylcholine receptor. Public domain.

At a synapse

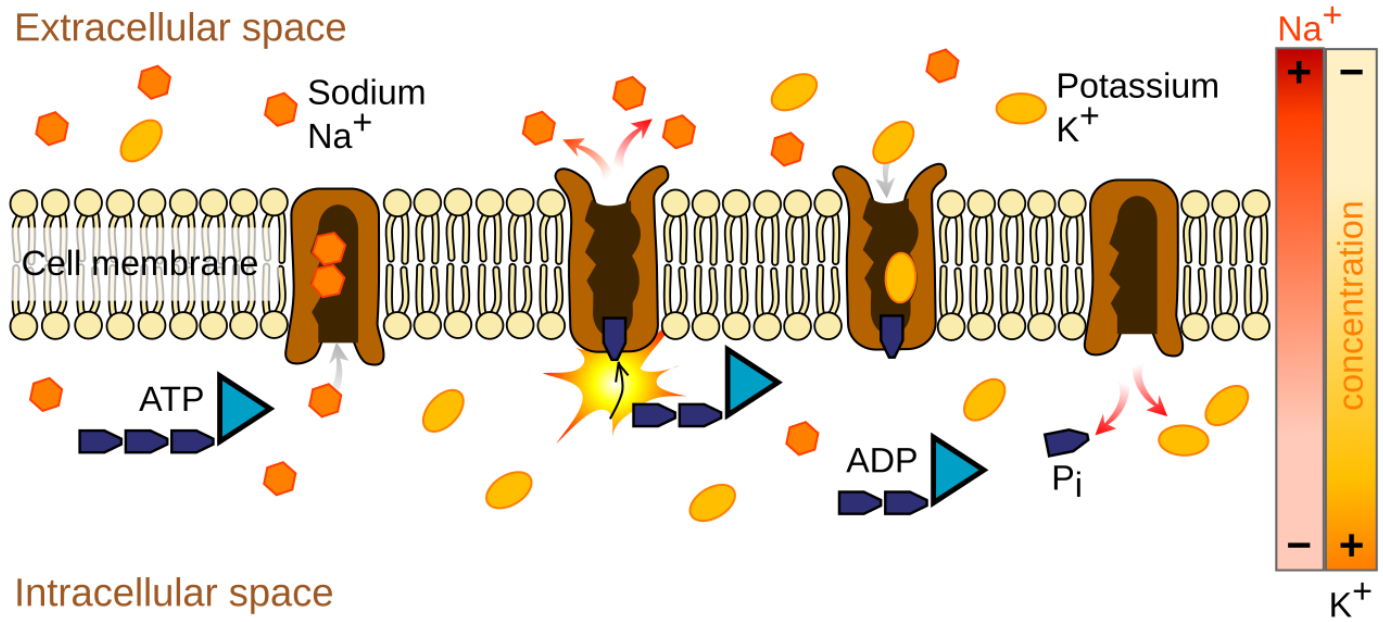
Acetylcholine binds, the receptor changes shape, the pore opens and Na^+ flows in. It closes when ACh leaves.

Along the axon

When the inside becomes positive, voltage-gated K^+ channels open and K^+ diffuses out.

The sodium-potassium pump

B2.1.15



Sodium-potassium pump. M. Ruiz Villarreal, public domain.

Why it matters

Na^+ out and K^+ in, both against gradients, using ATP: gradients for nerves and cotransport.

1. 3 Na^+ bind inside

2. ATP phosphorylates the pump

3. Opens out: 3 Na^+ released

4. 2 K^+ bind; phosphate leaves

5. Opens in: 2 K^+ released

Practice: channels and pumps

B2.1.14–15

Compare

Voltage-gated and ligand-gated channels.

Calculate

Net charge moved out per pump cycle.

Explain

Why the pump is active transport but channels are not.

Use

facilitated diffusion, phosphorylation, gradient, conformation

Check-in: class 2

B2.1.14–15

1. State what opens a voltage-gated channel.
2. Give an example of a neurotransmitter-gated channel.
3. List the steps of the Na/K pump.
4. State the ratio of ions moved.
5. Explain why channels need no ATP.

How to use this

Answer without your notes, then check against slides 12–16.

Indirect active transport of glucose

B2.1.16



Big question

The pump uses ATP to keep Na^+ low inside; Na^+ flowing in drags glucose with it.

What if the pump stops?

B2.1.16

Ouabain

Blocks the pump: predict glucose uptake.

Cyanide

Stops ATP: why does glucose absorption stop?

Diabetes

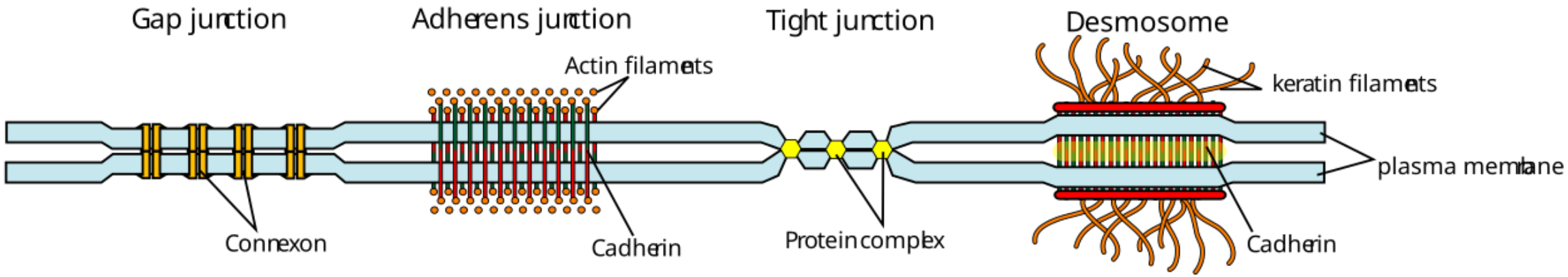
Why is glucose in urine when blood glucose is very high?

Link

C1.2 (ATP), C2.2 (impulses), D3.3 (kidney).

Cell adhesion: building tissues

B2.1.17



Cell junctions: gap, adherens, tight and desmosome. M. Ruiz Villarreal, public domain.

Junction	Role
Tight	Seals gaps between epithelial cells
Anchoring	Links cytoskeletons for strength
Gap	Channel joining cytoplasm of two cells

Cell-adhesion molecules

B2.1.17

CAMs are membrane proteins that bind CAMs on other cells or the matrix.

Cadherins: cell to cell. **Integrins**: cell to matrix.

Why it matters

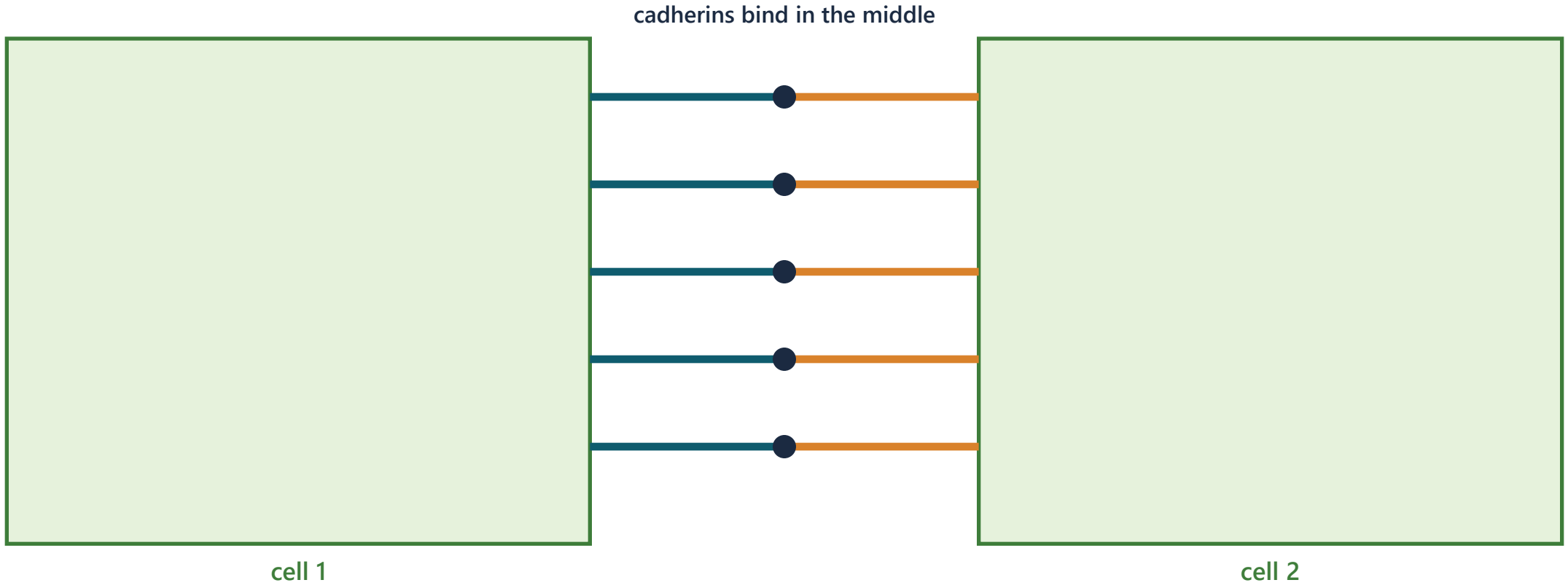
Cancer cells that lose cadherins can break away and spread (metastasis).

Plants

Few CAMs: walls interlock; plasmodesmata link cytoplasm.

Cadherins zip cells together

B2.1.17



Key idea

Matching CAMs on neighbouring cells hold tissues together.

Practice: cotransport and adhesion

B2.1.16–17

Explain

Why SGLT transport is indirect active transport.

Distinguish

Cadherins and integrins.

Suggest

Why tight junctions matter in the gut.

Use

gradient, cotransporter, cell-adhesion molecule, junction

Check-in: class 3

B2.1.16–17

1. Explain how glucose is absorbed against its gradient.
2. State why this is called indirect.
3. Name three kinds of cell junction.
4. Distinguish cadherins and integrins.
5. Explain how losing CAMs helps cancer spread.

How to use this

Answer without your notes, then check against slides 18–23.

Common mistakes to avoid

Not this	But this
The pump moves 3 K^+ in, 2 Na^+ out.	3 Na^+ out, 2 K^+ in.
Cotransport uses ATP directly.	It uses the Na^+ gradient.
Cholesterol only stiffens membranes.	It buffers fluidity both ways.
Channels pump ions.	Channels allow diffusion.

Summary: HL membranes

B2.1.11–17

	Key idea
B2.1.11	Unsaturated tails raise fluidity
B2.1.12	Cholesterol buffers fluidity
B2.1.13	Fluidity allows vesicles
B2.1.14	Voltage- and ligand-gated channels
B2.1.15	Na/K pump: 3 out, 2 in, ATP
B2.1.16	SGLT: indirect active transport
B2.1.17	CAMs and junctions

Check that you can...

1. **Explain** how fatty acids and cholesterol affect fluidity
2. **Outline** endocytosis and exocytosis
3. **Compare** gated channels
4. **Describe** the Na/K pump
5. **Explain** indirect active transport
6. **Outline** cell adhesion

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. State how unsaturated fatty acids affect membrane fluidity.
2. Outline one difference between a voltage-gated and a neurotransmitter-gated channel.
3. Explain why glucose uptake in the intestine depends on ATP.

Image credits

Endocytosis types.svg – Mariana Ruiz Villarreal LadyofHats – Public domain
Phospholipid TvanBrussel.jpg – Author not stated – Copyrighted free use
Lake sturgeon (Acipenser fulvescens).png – Ryan Hagerty/USFWS – Public domain
Cholesterol molecule skeletal.png – Jynto (talk) – CC0
Exocytosis types.svg – Mariana Ruiz LadyofHats – Public domain
Nicotinic Acetylcholine receptor.png – S. Jähnichen assumed (b – Public domain
Scheme sodium-potassium pump-en.svg – LadyofHats Mariana Ruiz Villarreal – Public domain
Cell junction simplified en.svg – Mariana Ruiz LadyofHats – Public domain

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